

Multiple Choice (10 marks)

1. compute the line integral of $\int_K xy \, dx$, $\int_L xy \, dx$, where K is a straight line from (0,0) to (1,1) and L is the Parabola $y=x^2$ from (0,0) to (1,1). return the answer as a list

0.333, 0.2

0.1, 0.05

0.4, 0.3

0.5, 0.75

0.4, 0.5

2. For which of the following rings is it possible for the product of two nonzero elements to be zero?

(a) The ring of complex numbers with rational coefficients

(b) The ring of integers modulo 11

(c) The ring of continuous real-valued functions on $[0, 1]$

(d) The ring of rational numbers

(e) The ring of complex numbers

3. A team of volunteers collected a total of 5,144 *selling T-shirts at a charity concert. Each T-shirt was sold for* 8. What was the total number of T-shirts the volunteers sold?

(a) 650

(b) 700

(c) 680

(d) 655

(e) 620

4. Write 8 and 2 over 4 as an improper fraction in simplest form.

(a) 17 over 2

(b) 34 over 4

(c) 16 over 2

(d) 19 over 2

(e) 33 over 4

5. Identify a solution for the equation $5x - 5 = -10$ from the given numbers.

- (a) -2
- (b) -5
- (c) -4
- (d) 0
- (e) 2

True / False (10 marks)

Answer True or False

6. True or False: The covariance $\text{Cov}(X(s=1/2), X(t=1))$ for standard Brownian motion is equal to 1.6180.
7. True or False: The probability that all three mutually independent events A, B, and C occur is 0.36.
8. True or False: The total volume of water in 8 fish tanks, each containing 40 liters, is 400 liters.
9. True or False: An outlier is the mode of the data set, representing the most frequently occurring observation.
10. True or False: The correct answer to 'Consider the initial value problem $5u'' + 2u' + 7u = 0$, $u(0) = 2$, $u'(0) = 1$ Find the smallest...' is '14.5115'.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the process of determining the units digit of the number 7 raised to the power of 25. Discuss the underlying patterns and mathematical principles involved in your reasoning.
12. Evaluate the integral $\frac{2}{\pi} \int_0^{+\infty} \frac{\sin^2 t}{t^2} \cos(2xt) dt$ for $x = \frac{\sqrt{17}}{17}$, and discuss your findings, rounding the result to the hundredths place.

End of Paper